📝 Weekly Report – CogniStream Project (Week 1)

Project: CogniStream – Developer Flow-State & Cognitive Load Analytics

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Week-1 deliverables completed successfully ✅

1. Objective

To build the ingestion pipeline for developer activity data (GitHub, Slack, IDE) and automate daily runs using Apache Airflow. This pipeline forms the foundation for analysing developer flow-state and context-switching friction.

2. Work Completed

- **Day 1** ✅
  - Wrote Python extraction script for mock **GitHub API**.
  - Validated commit data ingestion into CSV format.
- **Day 2** ✅
  - Developed Python extraction script for mock **Slack API**.
  - Captured message events and stored them in structured output.
- **Day 3** ✅
  - Built Python extraction script for mock **IDE Activity API**.
  - Logged coding session events and saved them for analysis.
- **Day 4** ✅
  - Integrated all three scripts (GitHub, Slack, IDE).
  - Tested end-to-end data extraction pipeline.
  - Verified outputs in Data/Raw folder (CSV files generated correctly).
- **Day 5** ✅

    - Set up **Apache Airflow DAGs** for scheduling daily runs.
    - Configured orchestration to trigger ingestion scripts automatically.
- **Day 6** ✅
    - Validated Airflow runs and fixed minor issues.
    - Ensured logs and CSV outputs updated successfully.
- **Day 7** ✅
    - Conducted final review of ingestion pipeline.
    - Prepared documentation and weekly report summarizing progress.

3. Outputs ✅

- **Scripts:**
    - python_github_ingestion.py
    - python_slack_ingestion.py
    - python_ide_ingestion.py
    - Integrated DAG: cognistream_ingestion_dag.py
- **Data Files:**
    - Data/Raw/github_commits.csv
    - Data/Raw/slack_data.csv
    - Data/Raw/ide_activity.csv
- **Logs:**
    - run_log.txt (execution proof of daily runs)

4. Validation ✅

- All ingestion scripts executed successfully.
- Airflow DAGs scheduled and tested.
- Logs recorded with timestamps.

- CSV files updated with latest ingested data.

5. Next Steps (Week 2)

- Deploy **ClickHouse** database.
- Use **Polars** to clean and transform ingested JSON payloads.
- Load processed data into ClickHouse event tables.
- Begin building base metrics for dashboard (commits, IDE hours).